CPU Scheduling Algorithms - Complete Placement Notes

Topics

Algorithms: FCFS, SJF, SRTF, Priority Scheduling, Round Robin.
Metrics: Arrival Time, Burst Time, Completion Time, Waiting Time, Turnaround Time, Response Time.
Gantt Charts, Preemptive vs Non-preemptive.

Formulas

TAT = CT - AT
WT = TAT - BT
RT = First Start Time - AT

Algorithm Summary

FCFS: Non-preemptive, executes by arrival order. Pros: Simple, no starvation. Cons: Convoy effect.

SJF: Non-preemptive, shortest burst first. Minimum average waiting time. Can starve long jobs.

SRTF: Preemptive SJF. Shortest remaining time runs. Better response, more context switches.

Priority: Highest priority first. Can be preemptive/non-preemptive. Starvation solved by Aging.

Round Robin: Preemptive, fixed time quantum. Fair, ideal for time-sharing.

Preemptive vs Non-preemptive

Preemptive: CPU can interrupt a running process (SRTF, RR, Preemptive Priority). Better response but more context switches.
Non-preemptive: Process runs until completion/block (FCFS, SJF). Lower overhead.

Most Asked Interview Questions

1. What is CPU scheduling?
2. Why is CPU scheduling needed?
3. Explain FCFS with example.
4. What is Convoy Effect?
5. Why does SJF minimize average waiting time?
6. Difference between SJF and SRTF?
7. What is starvation? How does aging solve it?
8. Explain Priority Scheduling.
9. Why is Round Robin used in time-sharing systems?
10. What happens if time quantum is too small or too large?
11. Difference between preemptive and non-preemptive scheduling?
12. Define Waiting Time, Turnaround Time, Response Time.
13. Draw a Gantt chart and calculate CT, TAT, WT, RT.
14. Which algorithm is used in modern operating systems?

Rapid Fire

Simplest: FCFS
Minimum waiting time: SJF
Interactive systems: Round Robin
Starvation: SJF, SRTF, Priority
Time quantum: Round Robin

Placement Tips

Practice numerical problems by drawing Gantt charts and calculating CT, TAT, WT and RT for FCFS, SJF, SRTF, Priority and Round Robin. Interviewers frequently compare algorithms and ask for the best choice in different scenarios.